

Differential Equations

ANSWER KEY



Solving by separation of variables

- 1 Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$.
Separate: $\frac{dy}{y} = \frac{dx}{x}$. Integrate: $\ln|y| = \ln|x| + C$, so $y = Ax$. Using $y = 6, x = 2$: $A = 3$. Solution: $y = 3x$.
- 2 Solve $\frac{dy}{dx} = 3x^2y$, given $y = 2$ when $x = 0$.
Separate: $\frac{dy}{y} = 3x^2 dx$. Integrate: $\ln|y| = x^3 + C$, so $y = Ae^{x^3}$. Using $y = 2, x = 0$: $A = 2$. Solution: $y = 2e^{x^3}$.
- 3 Solve $\frac{dy}{dx} = \frac{2x}{y^2}$, given $y = 2$ when $x = 0$.
Separate: $y^2 dy = 2x dx$. Integrate: $\frac{y^3}{3} = x^2 + C$. Using $y = 2, x = 0$: $C = \frac{8}{3}$. So $y^3 = 3x^2 + 8$, giving $y = (3x^2 + 8)^{1/3}$.
- 4 Solve $\frac{dy}{dx} = \frac{y^2}{1-x}$ for $x > 0$, given $y = 1$ when $x = 1$.
Separate: $\frac{dy}{y^2} = \frac{dx}{1-x}$. Integrate: $-\frac{1}{y} = \ln|x| + C$. Using $y = 1, x = 1$: $C = -1$. So $-\frac{1}{y} = \ln x - 1$, giving $y = \frac{1}{1 - \ln x}$.

Forming differential equations from a description

- 5 The rate of decrease of a quantity Q is proportional to Q itself. Write a differential equation for $\frac{dQ}{dt}$, using $k > 0$ as the constant of proportionality.
 $\frac{dQ}{dt} = -kQ$.
- 6 A tank drains so that the rate of decrease of its volume V is proportional to $\sqrt{V}$. Write a differential equation for $\frac{dV}{dt}$, using $k > 0$.
 $\frac{dV}{dt} = -k\sqrt{V}$.
- 7 A chemical reaction's concentration C decreases at a rate proportional to the square of the concentration. Write a differential equation for $\frac{dC}{dt}$, using $k > 0$.
 $\frac{dC}{dt} = -kC^2$.

Solving modeling equations with initial conditions

- 8 A population P satisfies $\frac{dP}{dt} = 0.08P$. If $P = 500$ when $t = 0$, find P when $t = 10$, to the nearest whole number.
 $P(t) = 500e^{0.08t}$. $P(10) = 500e^{0.8} = 500(2.225541) \approx 1113$.
- 9 Solve $\frac{dV}{dt} = -k\sqrt{V}$ (from the tank problem above), given $V(0) = V_0$.
Separate: $V^{-1/2} dV = -k dt$. Integrate: $2\sqrt{V} = -kt + C$. Using $V(0) = V_0$: $C = 2\sqrt{V_0}$. So $\sqrt{V} = \sqrt{V_0} - \frac{kt}{2}$, giving $V(t) = \left(\sqrt{V_0} - \frac{kt}{2}\right)^2$.

First-order linear differential equations using an integrating factor

- 10 Solve $\frac{dy}{dx} + 3y = 12$ using an integrating factor, given $y = 2$ when $x = 0$.
Integrating factor: e^{3x} . $\frac{d}{dx}(ye^{3x}) = 12e^{3x}$. Integrate: $ye^{3x} = 4e^{3x} + C$, so $y = 4 + Ce^{-3x}$. Using $y = 2, x = 0$: $C = -2$. Solution: $y = 4 - 2e^{-3x}$.
- 11 Solve $\frac{dy}{dx} - 2y = e^{3x}$ using an integrating factor, given $y = 1$ when $x = 0$.
Integrating factor: e^{-2x} . $\frac{d}{dx}(ye^{-2x}) = e^{3x}e^{-2x} = e^x$. Integrate: $ye^{-2x} = e^x + C$, so $y = e^{3x} + Ce^{2x}$. Using $y = 1, x = 0$: $C = 0$. Solution: $y = e^{3x}$.
- 12 Solve $\frac{dy}{dx} + y = 5$ using an integrating factor, given $y = 0$ when $x = 0$.
Integrating factor: e^x . $\frac{d}{dx}(ye^x) = 5e^x$. Integrate: $ye^x = 5e^x + C$, so $y = 5 + Ce^{-x}$. Using $y = 0, x = 0$: $C = -5$. Solution: $y = 5 - 5e^{-x}$.
- 13 Solve $\frac{dy}{dx} + 4y = 8$ using an integrating factor, given $y = 3$ when $x = 0$.
Integrating factor: e^{4x} . $\frac{d}{dx}(ye^{4x}) = 8e^{4x}$. Integrate: $ye^{4x} = 2e^{4x} + C$, so $y = 2 + Ce^{-4x}$. Using $y = 3, x = 0$: $C = 1$. Solution: $y = 2 + e^{-4x}$.
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Further integrating factor examples

- 14 Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$ for $x > 0$ using an integrating factor, given $y = 1$ when $x = 1$.
Integrating factor: $e^{\int \frac{1}{x} dx} = x$. $\frac{d}{dx}(xy) = x^3$. Integrate: $xy = \frac{x^4}{4} + C$, so $y = \frac{x^3}{4} + \frac{C}{x}$. Using $y = 1, x = 1$: $C = \frac{3}{4}$. Solution: $y = \frac{x^3}{4} + \frac{3}{4x}$.
- 15 Solve $\frac{dy}{dx} - \frac{2y}{x} = x^3$ for $x > 0$ using an integrating factor, given $y = 3$ when $x = 1$.
Integrating factor: $e^{\int -\frac{2}{x} dx} = x^{-2}$. $\frac{d}{dx}(yx^{-2}) = x$. Integrate: $yx^{-2} = \frac{x^2}{2} + C$, so $y = \frac{x^4}{2} + Cx^2$. Using $y = 3, x = 1$: $C = \frac{5}{2}$. Solution: $y = \frac{x^4}{2} + \frac{5}{2}x^2$.
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Verifying solutions to differential equations

- 16 Verify that $y = 3e^{2x}$ satisfies $\frac{dy}{dx} - 2y = 0$.
 $\frac{dy}{dx} = 6e^{2x}$. Substituting: $6e^{2x} - 2(3e^{2x}) = 6e^{2x} - 6e^{2x} = 0$, confirming the equation is satisfied.
- 17 Verify that $y = x^2 + Ce^{-x}$ is a general solution of $\frac{dy}{dx} + y = x^2 + 2x$.
 $\frac{dy}{dx} = 2x - Ce^{-x}$. Substituting: $(2x - Ce^{-x}) + (x^2 + Ce^{-x}) = 2x + x^2$, which matches the right-hand side $x^2 + 2x$ exactly, confirming the equation is satisfied for any constant C .
- 18 Verify that $y = \frac{1}{2}\sin(2x)$ satisfies $\frac{d^2y}{dx^2} + 4y = 0$.
 $\frac{dy}{dx} = \cos(2x)$. $\frac{d^2y}{dx^2} = -2\sin(2x)$. Substituting:
 $-2\sin(2x) + 4\left(\frac{1}{2}\sin(2x)\right) = -2\sin(2x) + 2\sin(2x) = 0$, confirming the equation is satisfied.

General and particular solutions

- 19 Find the general solution of $\frac{dy}{dx} = 4x^3 - 6x$, and explain what determines a particular solution from this family.

Integrating directly: $y = x^4 - 3x^2 + C$. A particular solution is obtained by specifying one point that the curve must pass through, which fixes the value of the arbitrary constant C .

- 20 A curve satisfies $\frac{dy}{dx} = 6x^2 - 4$ and passes through $(1, -1)$. Find the equation of this particular solution.
 $y = 2x^3 - 4x + C$. Using $(1, -1)$: $-1 = 2 - 4 + C \Rightarrow C = 1$. Solution: $y = 2x^3 - 4x + 1$.

- 21 A curve satisfies $\frac{dy}{dx} = \frac{1}{x}$ for $x > 0$ and passes through $(1, 4)$. Find the particular solution.
 $y = \ln x + C$. Using $(1, 4)$: $4 = 0 + C \Rightarrow C = 4$. Solution: $y = \ln x + 4$.

Applications: Newton's law of cooling

- 22 A cup of tea at 85°C is placed in a room at 22°C . Using $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 8 minutes the tea has cooled to 65°C , find k , to three decimal places.

$$65 = 22 + 63e^{-8k} \Rightarrow e^{-8k} = \frac{43}{63} \approx 0.6825. \text{ Taking natural logs: } -8k \approx -0.3819, \text{ so } k \approx 0.0477.$$

- 23 Using the value of k found above, find the temperature of the tea after 20 minutes, to the nearest degree.

$$T(20) = 22 + 63e^{-0.0477(20)} = 22 + 63e^{-0.954} \approx 22 + 63(0.3852) \approx 22 + 24.27 \approx 46^\circ \text{C}.$$

- 24 A metal rod at 200°C cools in air at 25°C , with cooling constant $k = 0.03$ per minute. Find its temperature after 15 minutes, to the nearest degree.

$$T(15) = 25 + 175e^{-0.03(15)} = 25 + 175e^{-0.45} \approx 25 + 175(0.6376) \approx 25 + 111.59 \approx 137^\circ \text{C}.$$

A well-designed word problem: a mixing tank

- 25 This question has three parts. A tank contains 300 litres of brine with 10 kg of dissolved salt. Fresh water flows in at 6 litres/minute, and the well-mixed solution flows out at the same rate, keeping the volume constant. Let $S(t)$ be the mass of salt (kg) at time t minutes. (a) Explain why $\frac{dS}{dt} = -\frac{S}{50}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 10$. (c) Find how long it takes for the salt content to drop to 2 kg, to the nearest minute.

(a) Since the incoming water carries no salt, salt only leaves, at a rate equal to concentration times outflow: $\frac{S}{300} \times 6 = \frac{S}{50}$ kg/min, so $\frac{dS}{dt} = -\frac{S}{50}$. (b) Separate: $\frac{dS}{S} = -\frac{dt}{50}$. Integrate: $\ln S = -\frac{t}{50} + C$, so $S = Ae^{-t/50}$. Using $S(0) = 10$: $A = 10$. Solution: $S(t) = 10e^{-t/50}$. (c) $2 = 10e^{-t/50} \Rightarrow e^{-t/50} = 0.2 \Rightarrow -\frac{t}{50} = \ln 0.2 \approx -1.6094 \Rightarrow t \approx 80$ minutes.